LUCID-CT / ADAPTIVE VERIFIED DECISION SUPPORT

Clinical Imaging & Verification Report

Uncertainty-Calibrated & Adaptive Self-Verifying Pulmonary Assessment

REPORT
2026-10-19

PATIENT & SCAN PROFILE

PATIENT ID	SCAN DATE	SCAN FORMAT	SLICES
Lung_Dx-B0041	2026-10-19	Real DICOM Slice (C:\Users\spide\Desktop\LUCID-CT\1.dcm)	16

ESTIMATED CATEGORY

4A **Suspicious**

Calibrated Malignancy Risk: 49.4%

RISK TIER

INTERMEDIATE

Estimated diameter: 10.0 mm

ADAPTIVE VERIFICATION & RELIABILITY GATE

ADAPTIVELY_RESOLVED (CRI: 79.5%)

ADAPTIVELY_RESOLVED

Recommended Clinical Action

VERIFIED via Adaptive Re-Analysis: Initial inconsistency investigated and resolved with high multi-scale agreement (93.6%) and CRI (79.5%). Proceed with guideline-based clinical management.

TARGETED RE-ANALYSIS AUDIT TRAIL

Agreement with Baseline: 93.6%

* 1.25x High-Magnification Boundary Shell Zoom

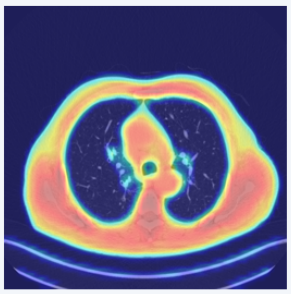
UNCERTAINTY & VERIFICATION INDICES

Epistemic Uncertainty	**0.0047**
Aleatoric Uncertainty	**0.6837**
Predictive Entropy	**0.6931**
Scan Quality Score	**73.9 / 100**
Grad-CAM Fidelity	**0.66 / 1.0**
Trait Physical Grounding	**0.68 / 1.0**

RADIOLOGICAL TRAIT EVIDENCE

Spiculation	**N/A**
Lobulation	**N/A**
Calcification	**N/A**
Subtlety	**N/A**
Margin	**N/A**
Sphericity	**N/A**

XAI LOCALIZATION & EVIDENCE GROUNDING



Estimated from diameter alone, assuming a baseline solid nodule. Nodule type, growth, and prior imaging were not available.